

Phase diagrams: examples

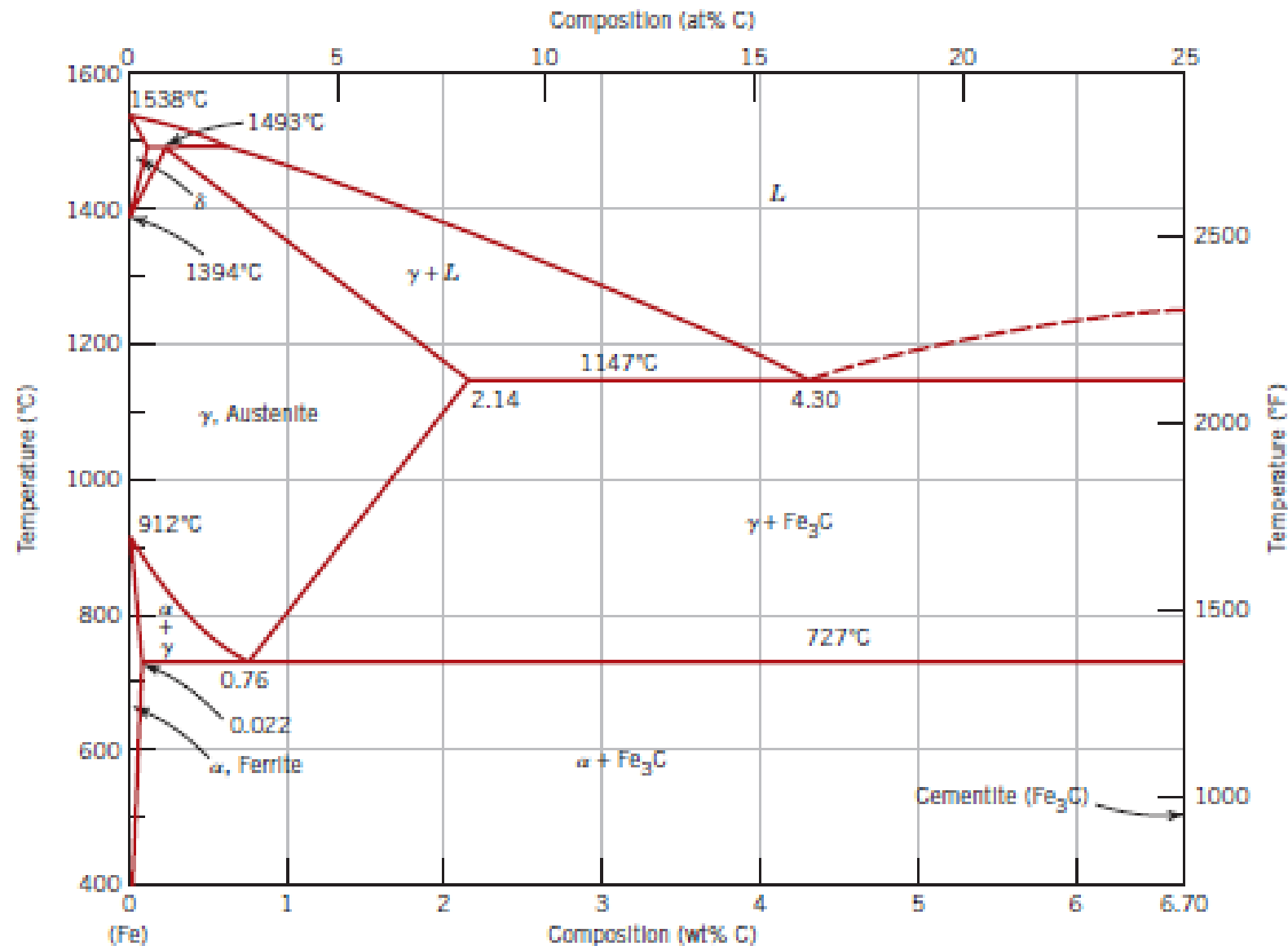


Figure 9.24 The iron–iron carbide phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

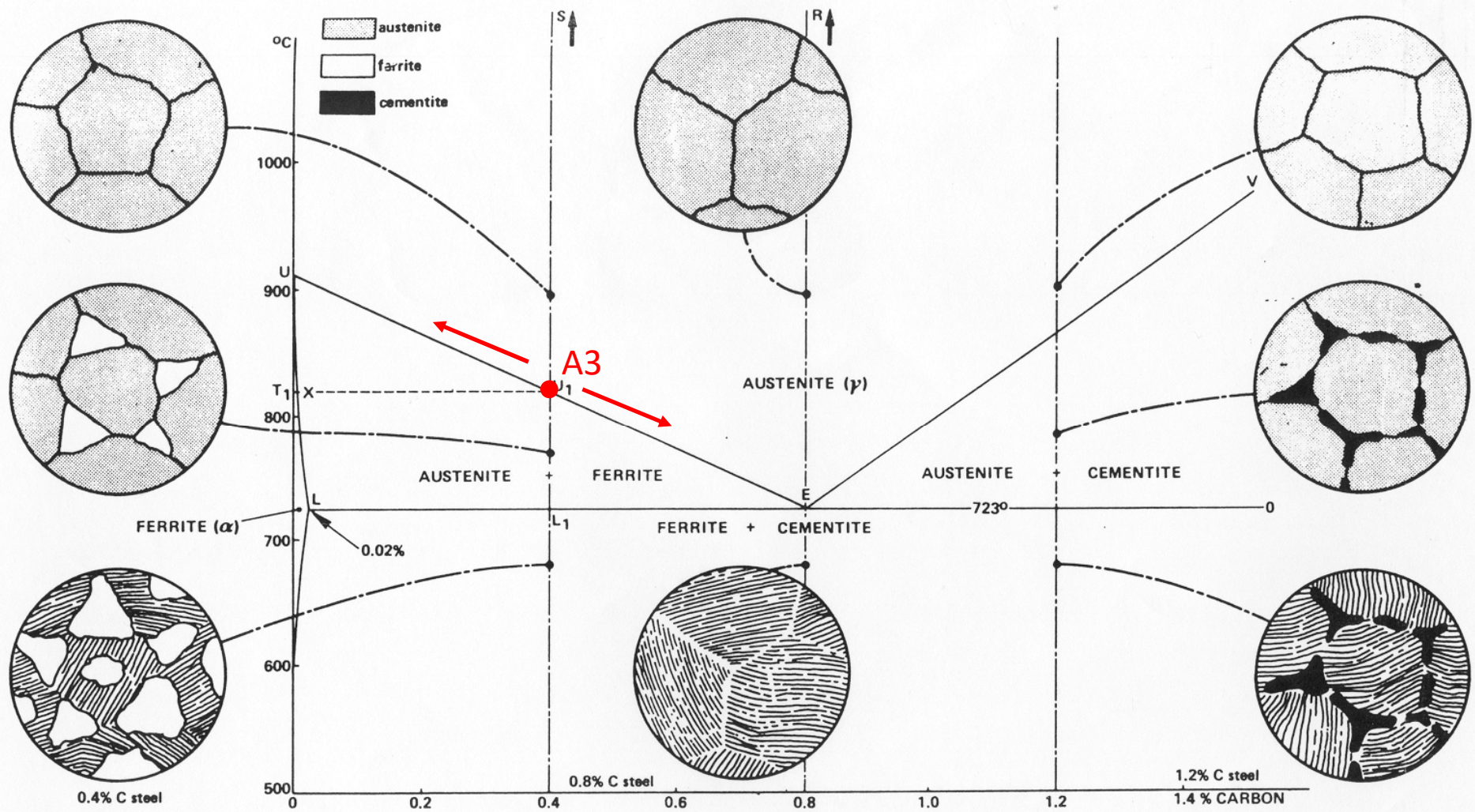
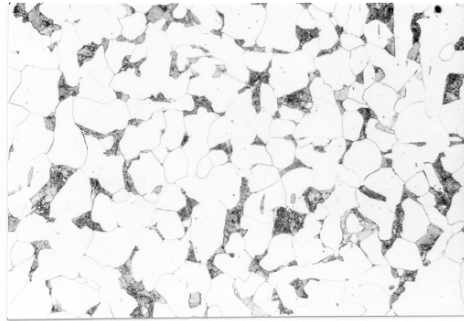


Fig. 9.3 The austenite→ferrite/cementite transformation in relation to the iron–carbon diagram.

Steel microstructure at room temperature

- Hypo-eutectoidic ($C < 0,8\%$): ferrite and pearlite

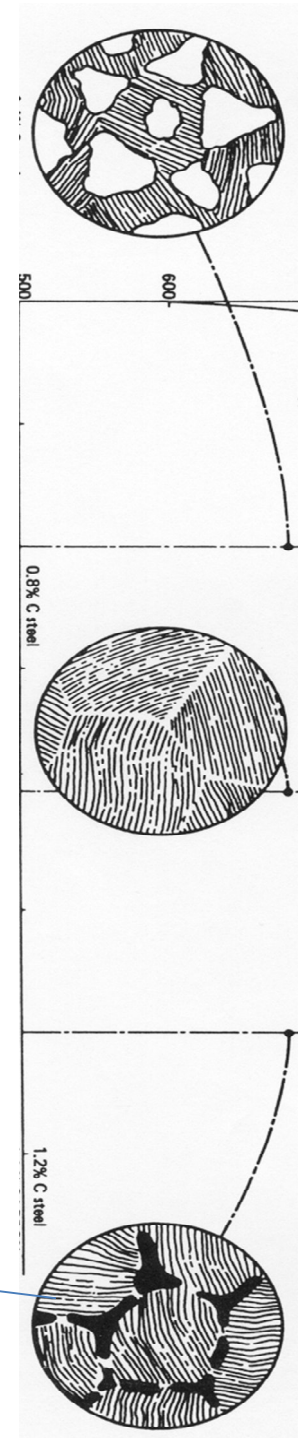


V = 350

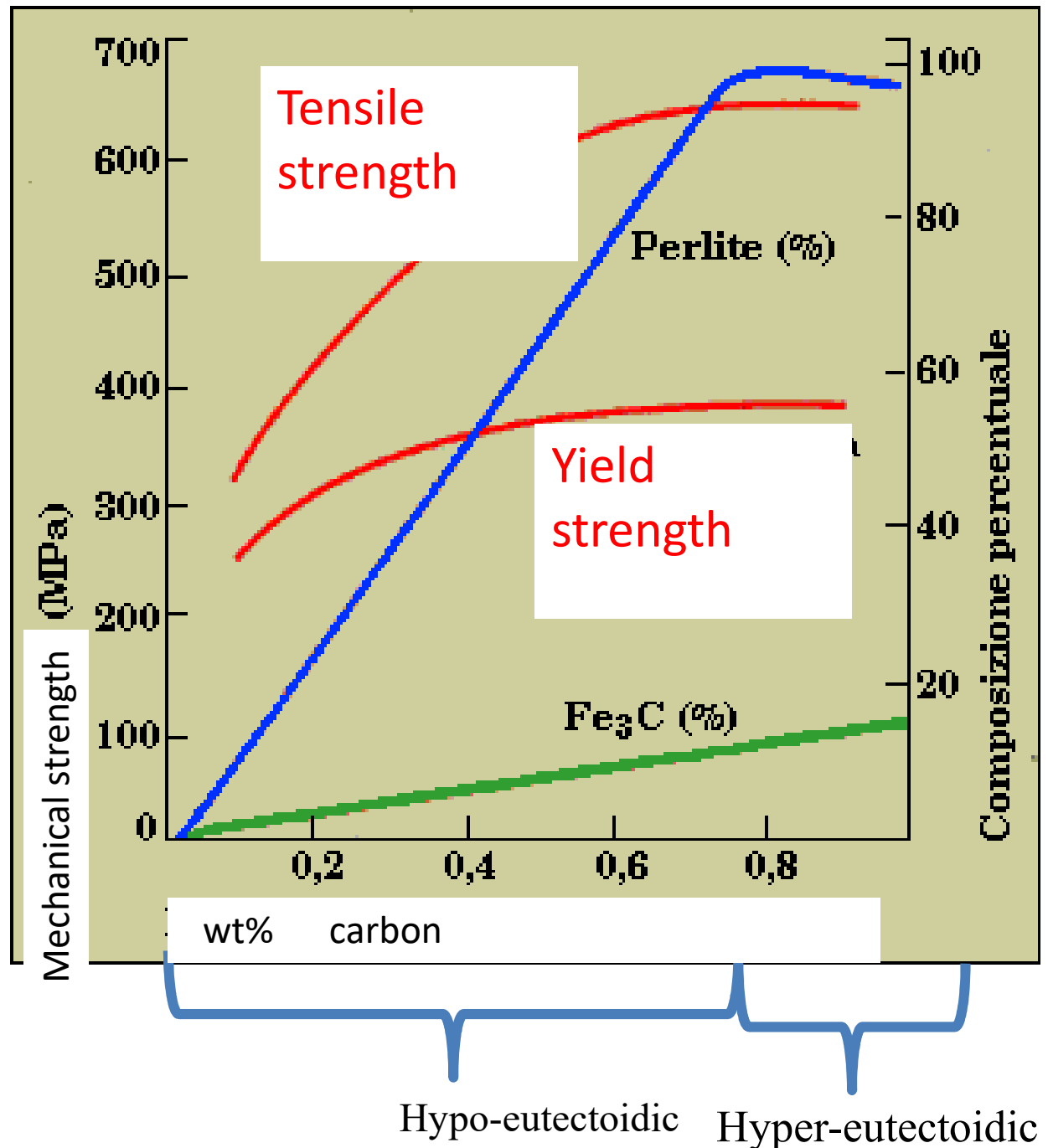
- eutectoidic ($C = 0,8\%$): pearlite



- Hyper-eutectoidic ($C > 0,8\%$): cementite and pearlite



Mechanical
properties of
steel
vs.
microstructure



Mechanical properties of steel vs. microstructure

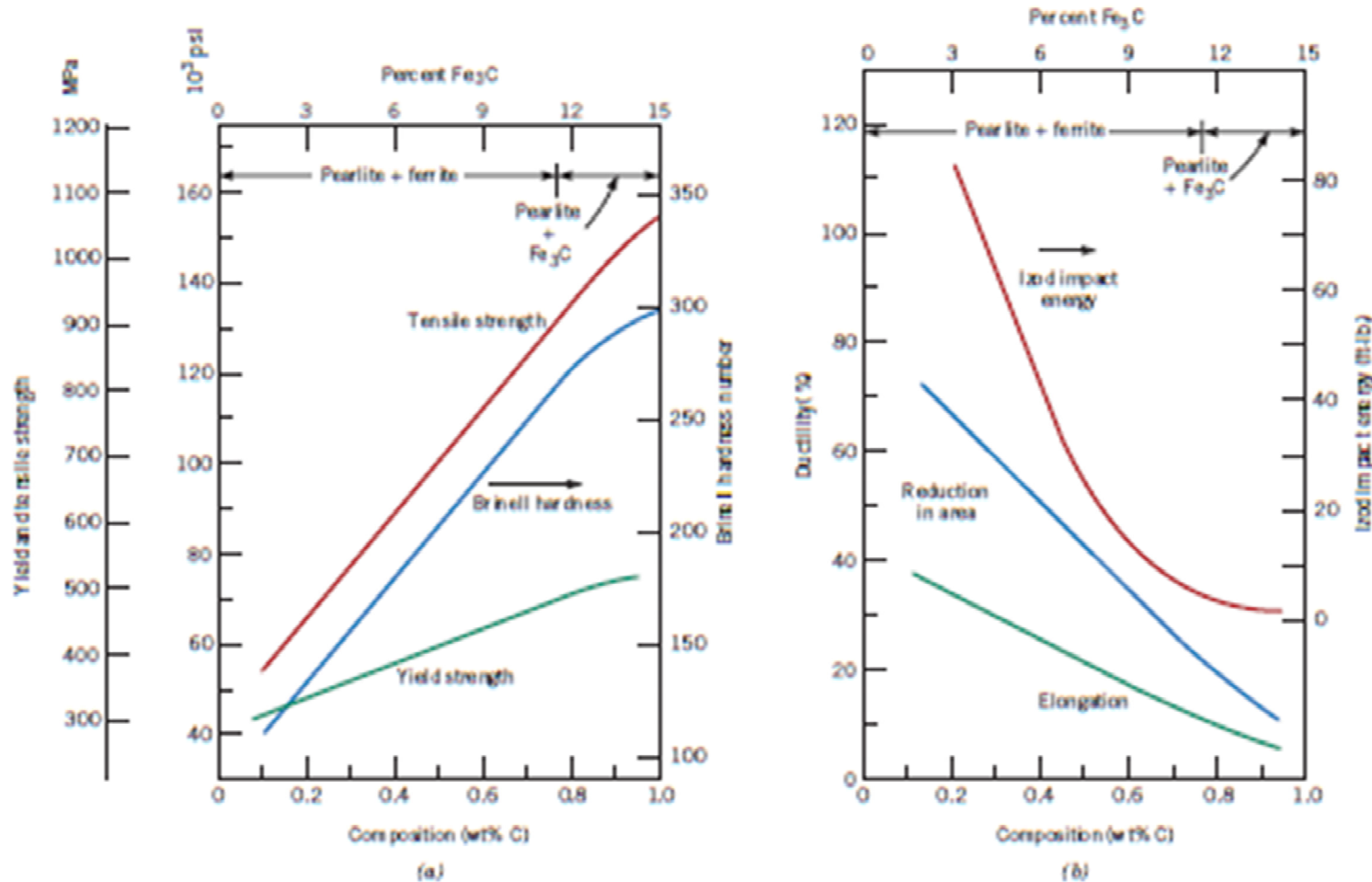
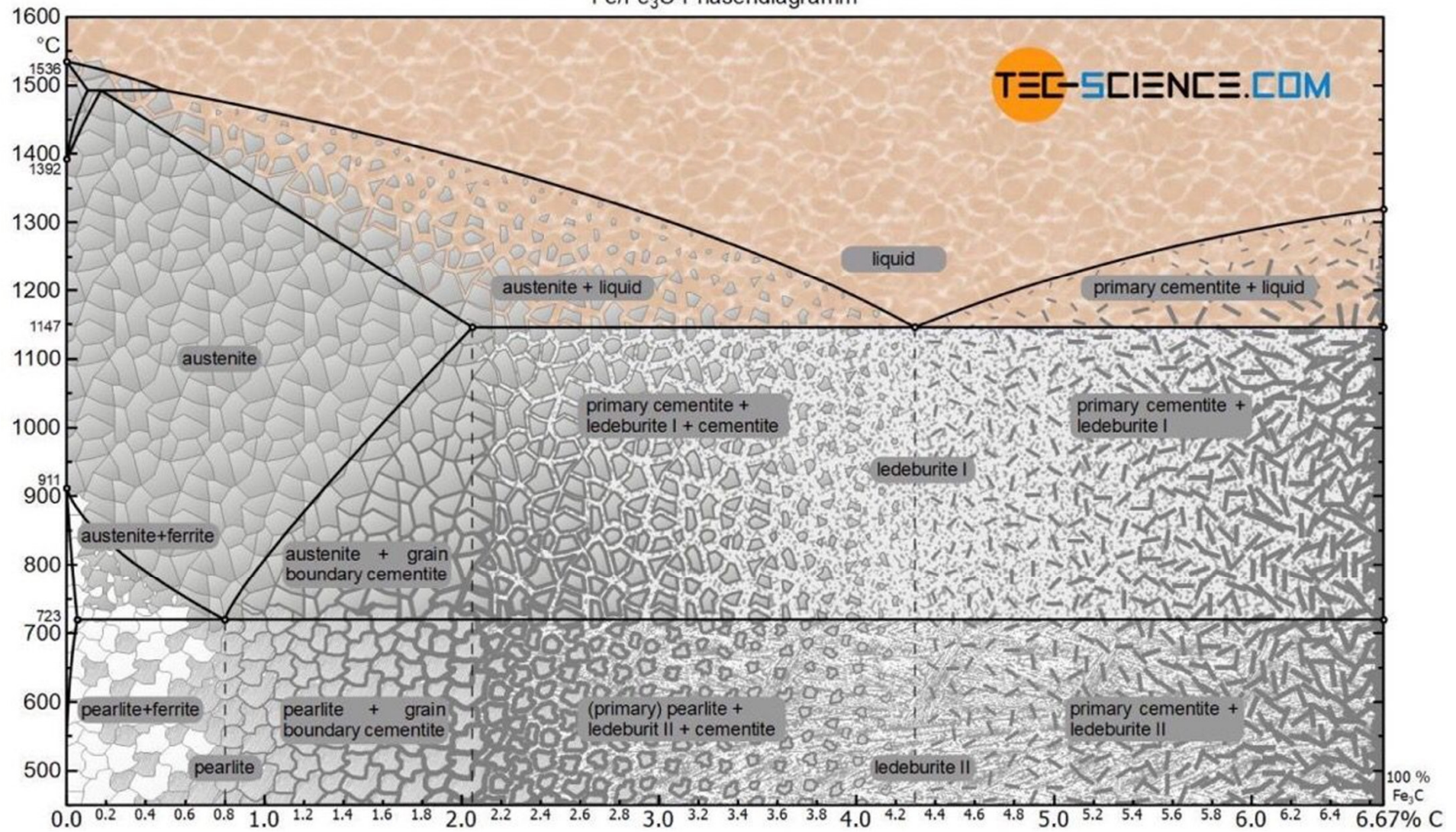


Figure 10.29 (a) Yield strength, tensile strength, and Brinell hardness versus carbon concentration for plain carbon steels having microstructures consisting of fine pearlite. (b) Ductility (%EL and %RA) and Izod impact energy versus carbon concentration for plain carbon steels having microstructures consisting of fine pearlite. [Data taken from *Metals Handbook: Heat Treating*, Vol. 4, 9th edition, V. Masseria (Managing Editor), American Society for Metals, 1981, p. 9.]

Fe/Fe₃C-Phasendiagramm



Microstructure of pearlite formed at different isothermal T:

If $> T \rightarrow$ coarse pearlite ($>$ diffusion)

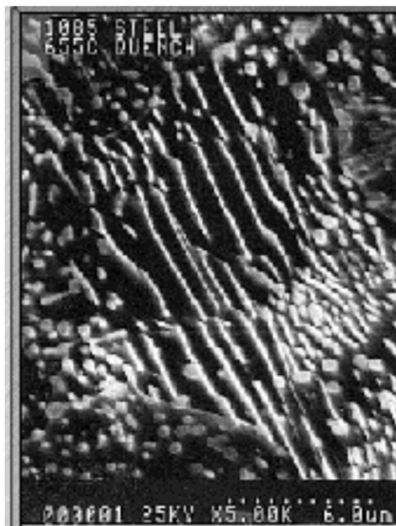
If $< T \rightarrow$ fine pearlite ($<$ diffusion)

655 °C

600 °C

534 °C

487 °C



(a)



(b)



(c)



(d)

Figure B.3.2a-b

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Figure B.3.2c-d

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If a steel with pearlitic microstructure is heated just below eutectoidic T for long time (e.g. 700 °C 18-24 h) → globular pearlite or *spheroidite*.

Driving force → minimizing the interfacial surface area between ferrite and cementite.

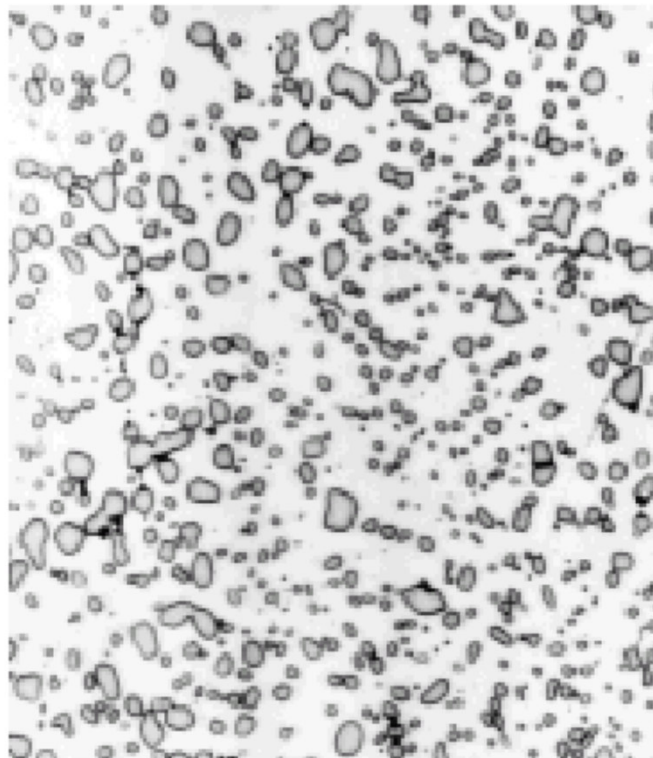


FIGURE 11.10 Photomicrograph of a steel having a spheroidite microstructure. The small particles are cementite; the continuous phase is α ferrite. 1000 $\times$. (Copyright 1971 by United States Steel Corporation.)

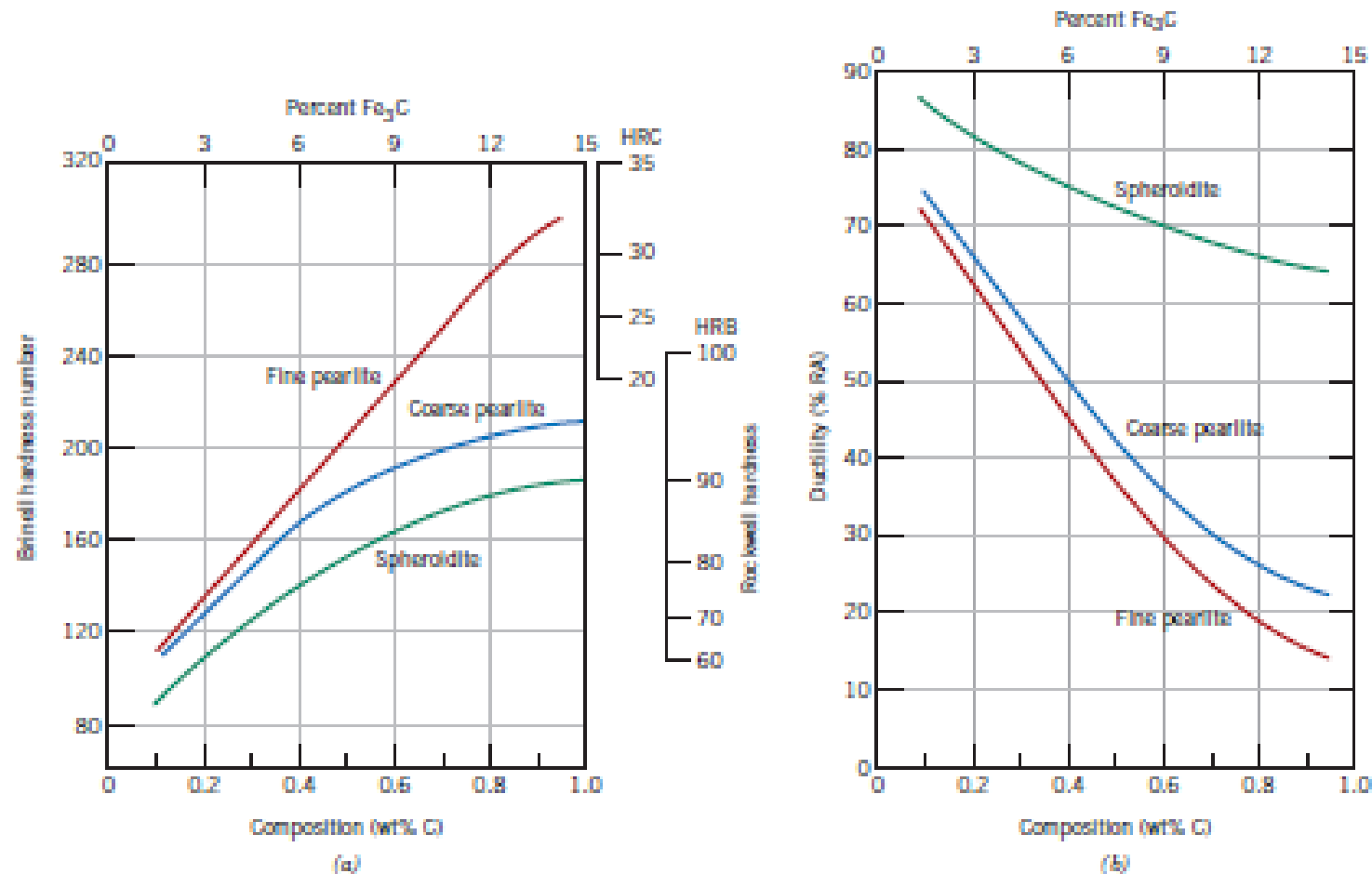


Figure 10.30 (a) Brinell and Rockwell hardness as a function of carbon concentration for plain carbon steels having fine and coarse pearlite as well as spheroidite microstructures. (b) Ductility (%RA) as a function of carbon concentration for plain carbon steels having fine and coarse pearlite as well as spheroidite microstructures. (Data taken from *Metals Handbook: Heat Treating*, Vol. 4, 9th edition, V. Masseria, Managing Editor, American Society for Metals, 1981, pp. 9 and 17.)

Spheroidite → < interfacial area → < density of GBs → easier plastic deformation

Martensite

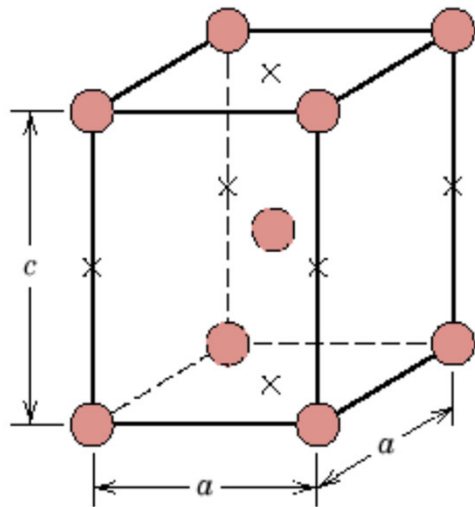
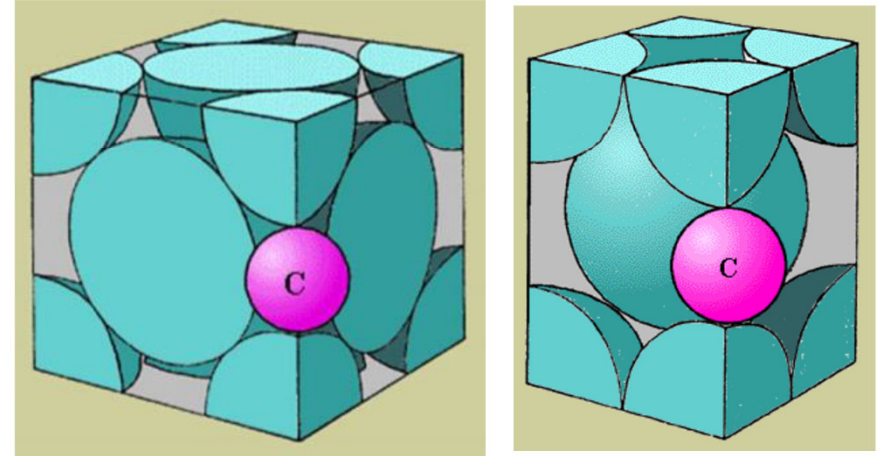


FIGURE 11.12 The body-centered tetragonal unit cell for martensitic steel showing iron atoms (circles) and sites that may be occupied by carbon atoms (crosses). For this tetragonal unit cell, $c > a$.



Austenite $\rightarrow$ Martensite



FIGURE 11.13 Photomicrograph showing the martensitic microstructure. The needle-shaped grains are the martensite phase, and the white regions are austenite that failed to transform during the rapid quench. 1220 $\times$. (Photomicrograph courtesy of United States Steel Corporation.)

Fast cooling from A3 to RT (quenching)

A3: point of austenitization (mobile point!)

Martensite: polymorphic transformation of austenite to martensite (single phase) $\rightarrow$ it can host a higher amount of C $\rightarrow$ solid solution effect against motion of dislocations.¹¹

Microstructure and mechanical properties of steels

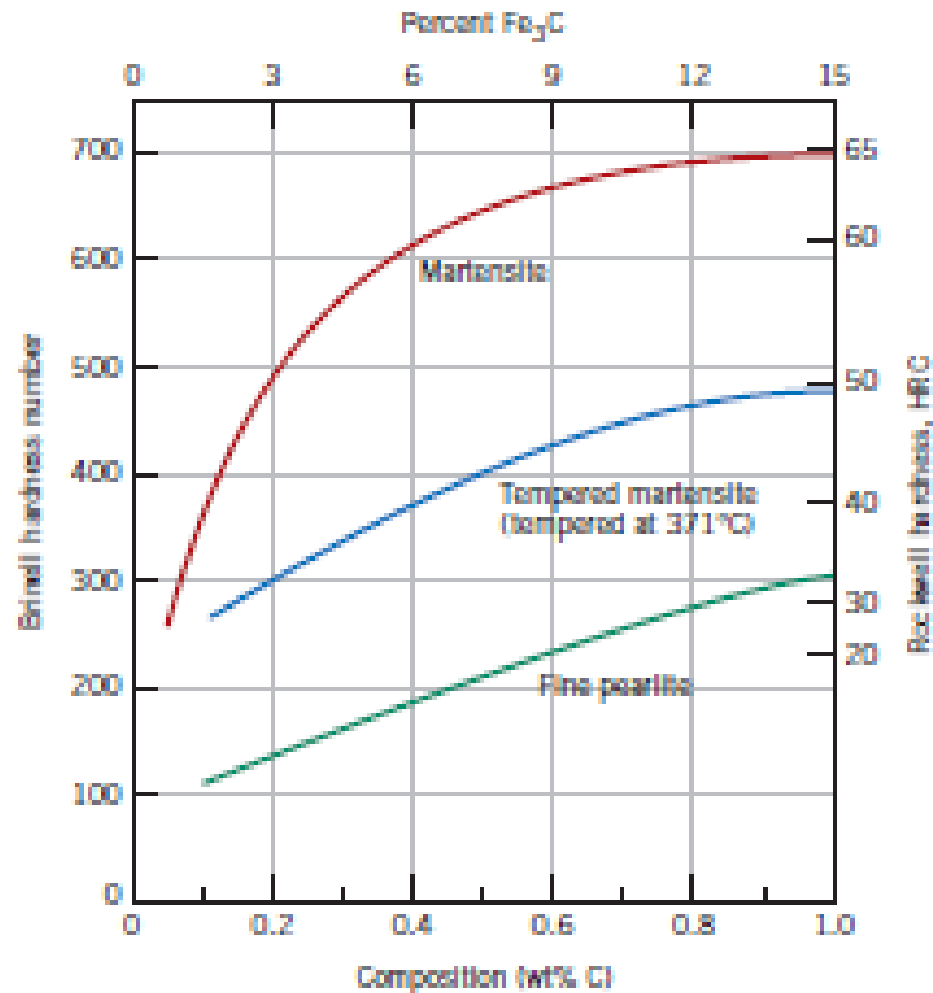
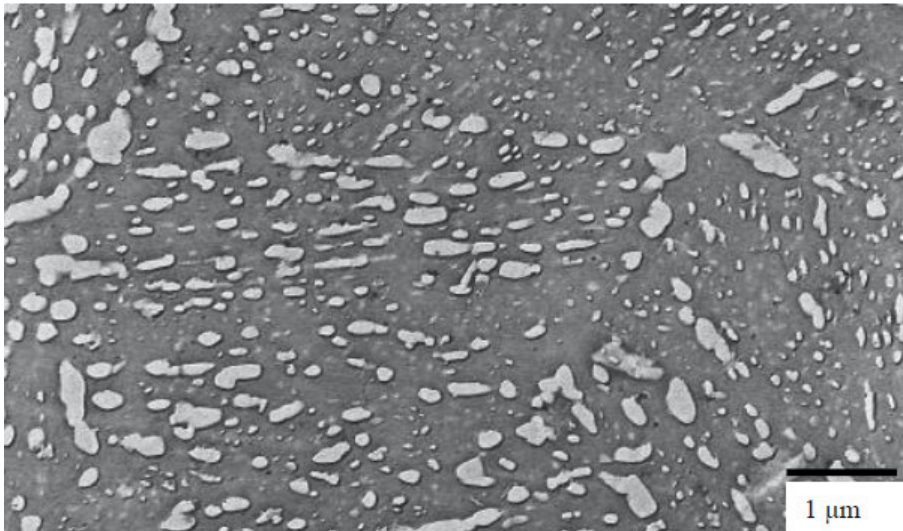


Figure 10.32 Hardness (at room temperature) as a function of carbon concentration for plain carbon martensitic, tempered martensitic [tempered at 371°C (700°F)], and pearlitic steels. (Adapted from Edgar C. Bain, *Functions of the Alloying Elements in Steel*, American Society for Metals, 1939, p. 36; and R. A. Grange, C. R. Hribal, and L. F. Porter, *Metall. Trans. A*, Vol. 8A, p. 1776.)

Tempering

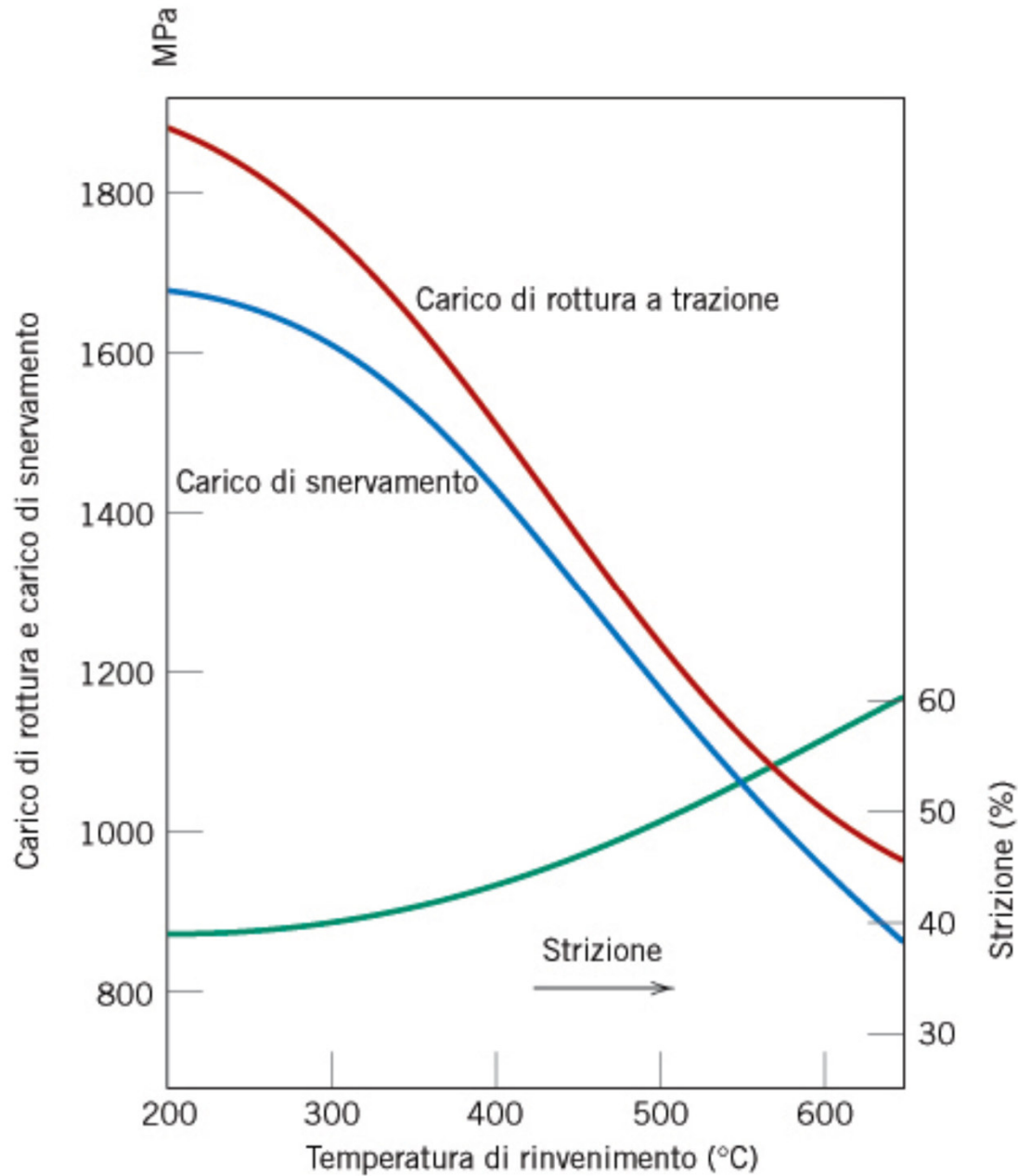
Increase of ductility and toughness of martensite → heat treatment at 250-650 °C



Phase transformation:

martensite → ferrite + cementite

Tempered martensite: fine particles of cementite in a matrix of ferrite
→ **sorbite** (similar to spheroidite)



For tempered martensite:

- > tempering $T \rightarrow$
- > rate of particle growth $\rightarrow$
- residual ferrite with low $C \rightarrow$
- > ductility but < yield strength and hardness

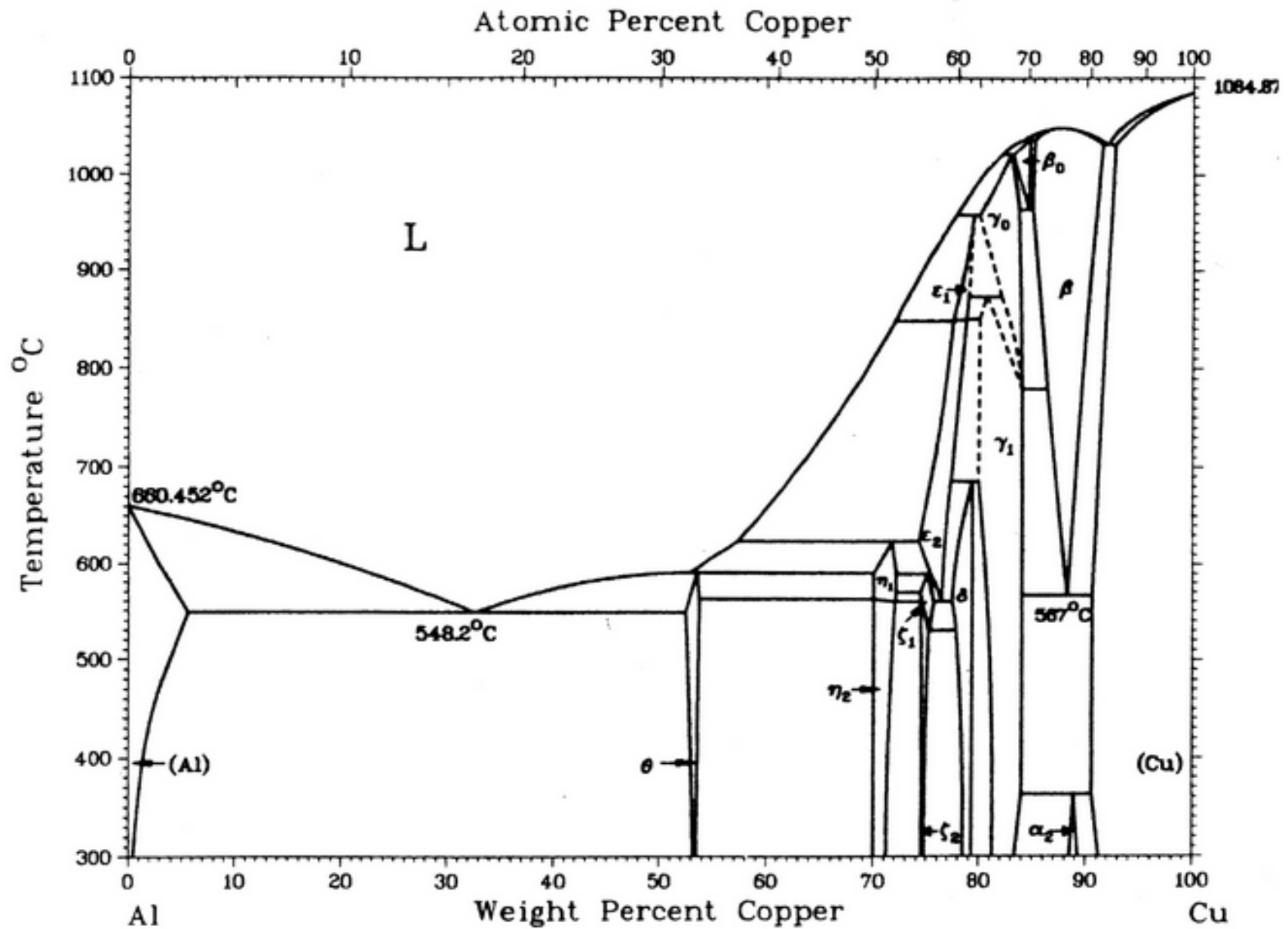


Heating to
complete
austenitization

quenching

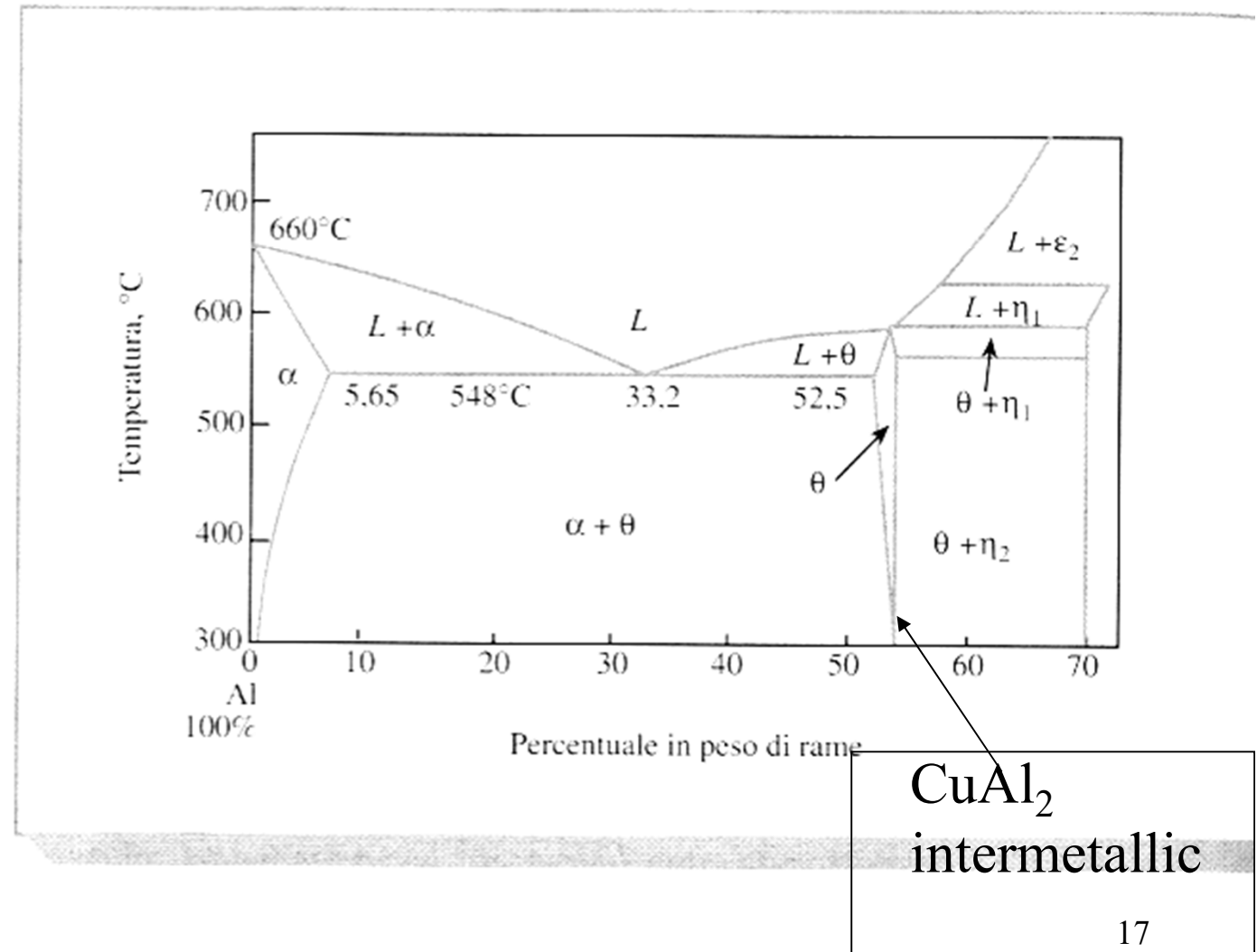
Increase
dislocations....

Al-Cu



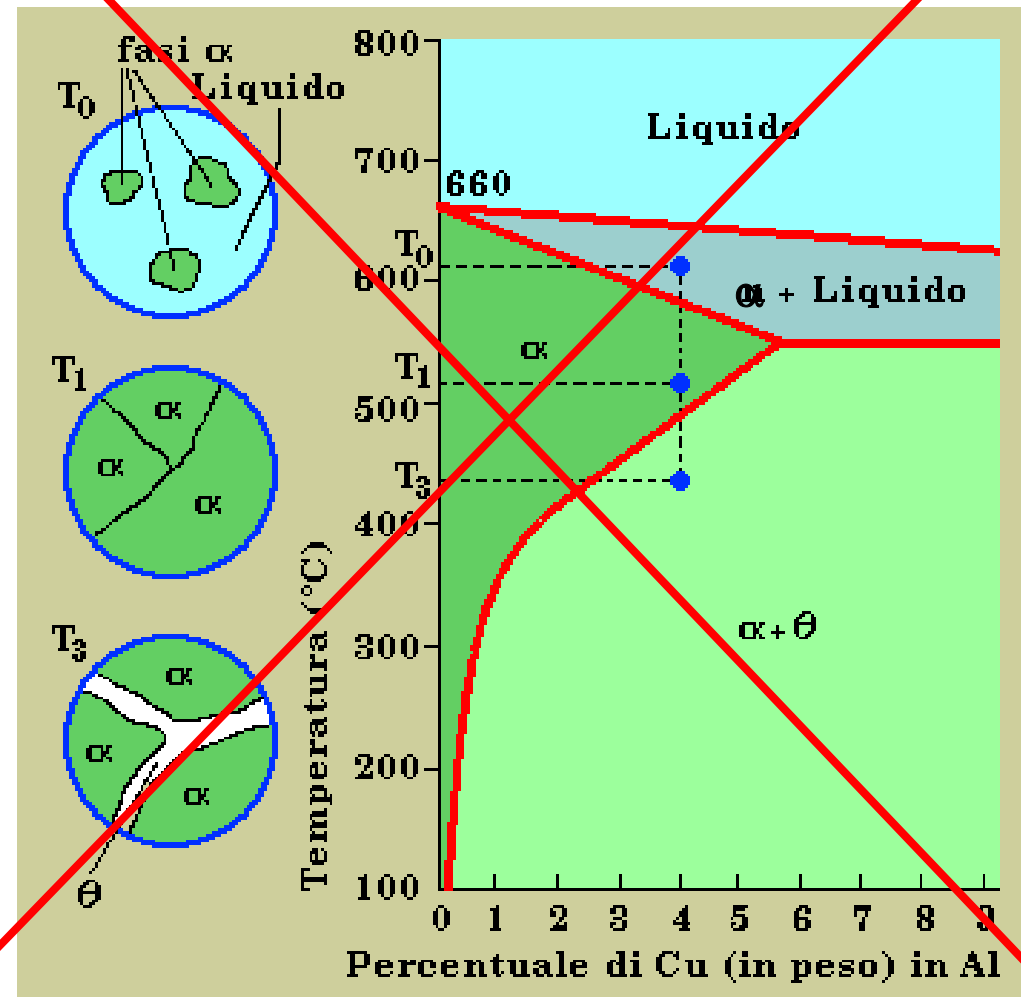
Al-Cu

Precipitation hardening for Al-Cu alloys



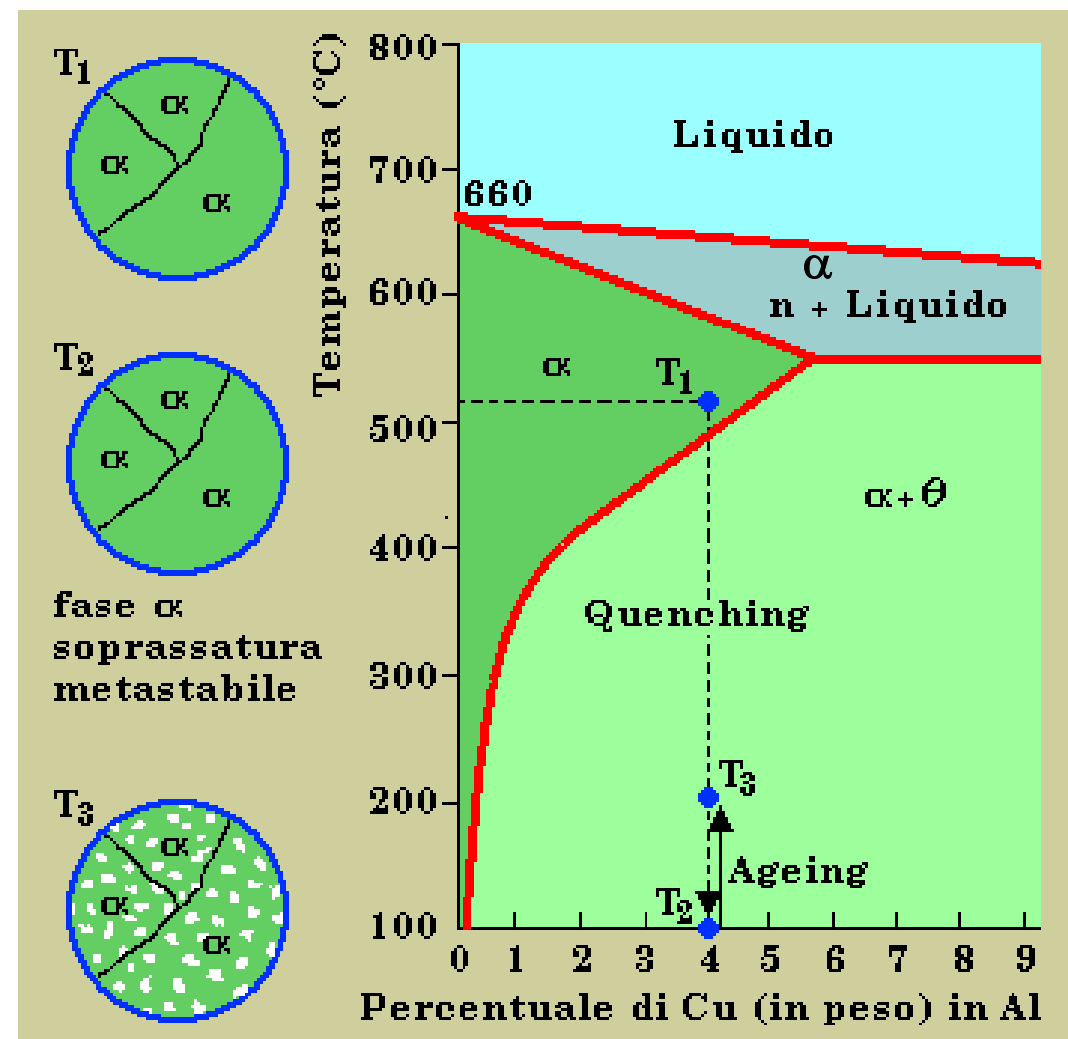
Al-Cu

Precipitation hardening for Al-Cu alloys



Al-Cu

Precipitation hardening for Al-Cu alloys



- Example of precipitation hardening: Al-Cu, Cu-Sn (bronze), Mg-Al,

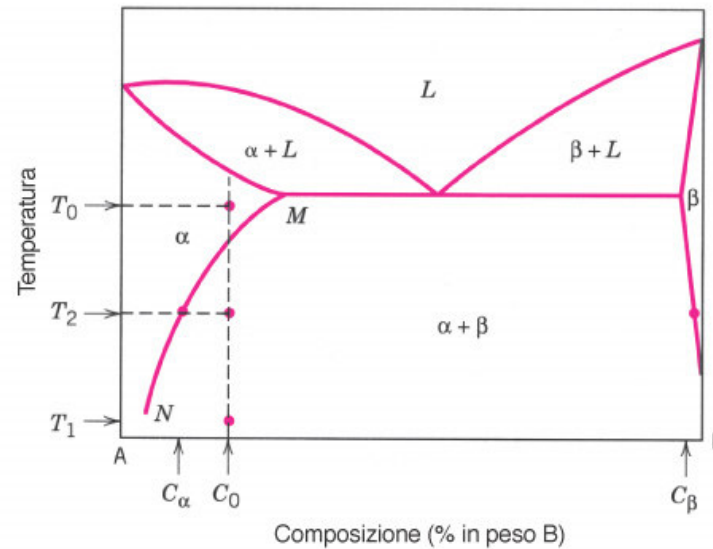


FIGURA 11.11 Diagramma di fase ipotetico per una lega indurente per precipitazione di composizione C_0 .

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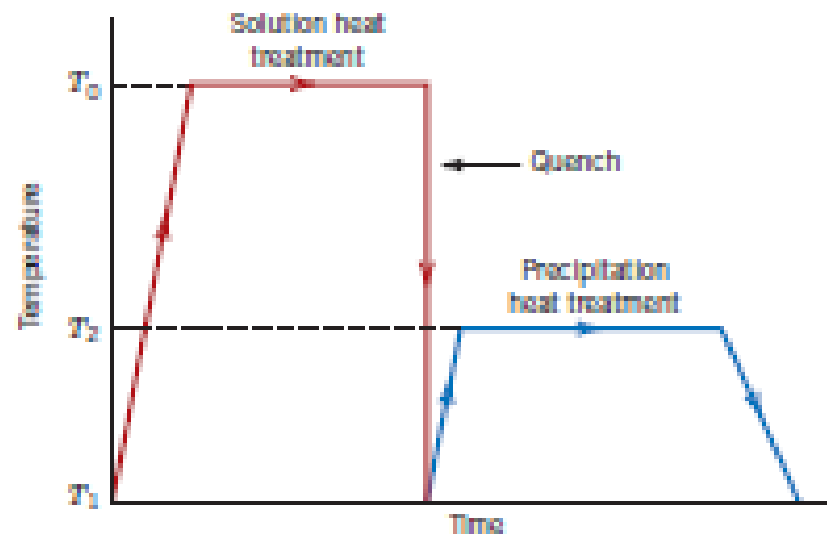


Figure 11.22 Schematic temperature-versus-time plot showing both solution and precipitation heat treatments for precipitation hardening.

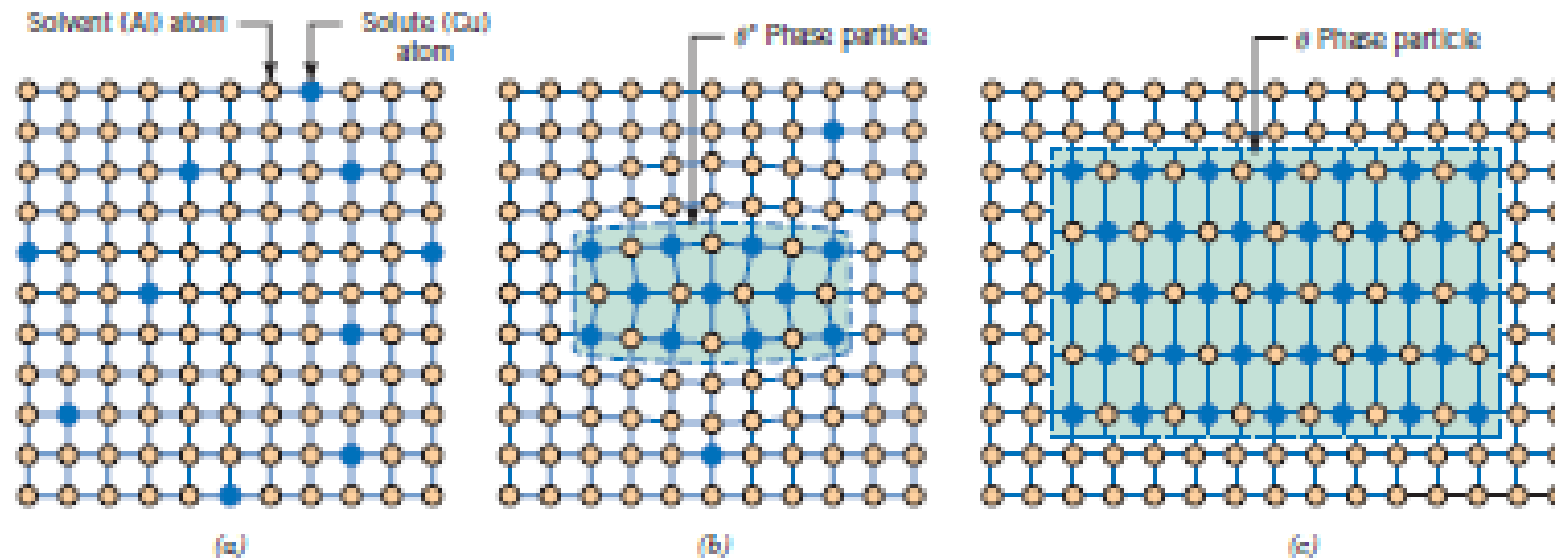
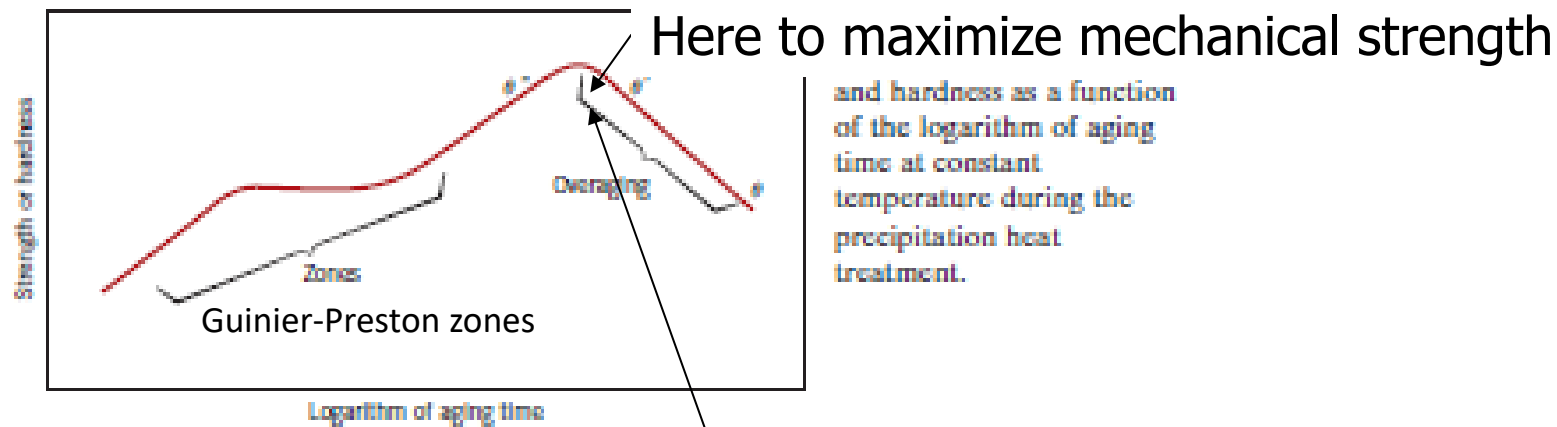


Figure 11.25 Schematic depiction of several stages in the formation of the equilibrium precipitate (θ) phase. (a) A supersaturated α solid solution. (b) A transition, θ' , precipitate phase. (c) The equilibrium θ phase, within the α -matrix phase.

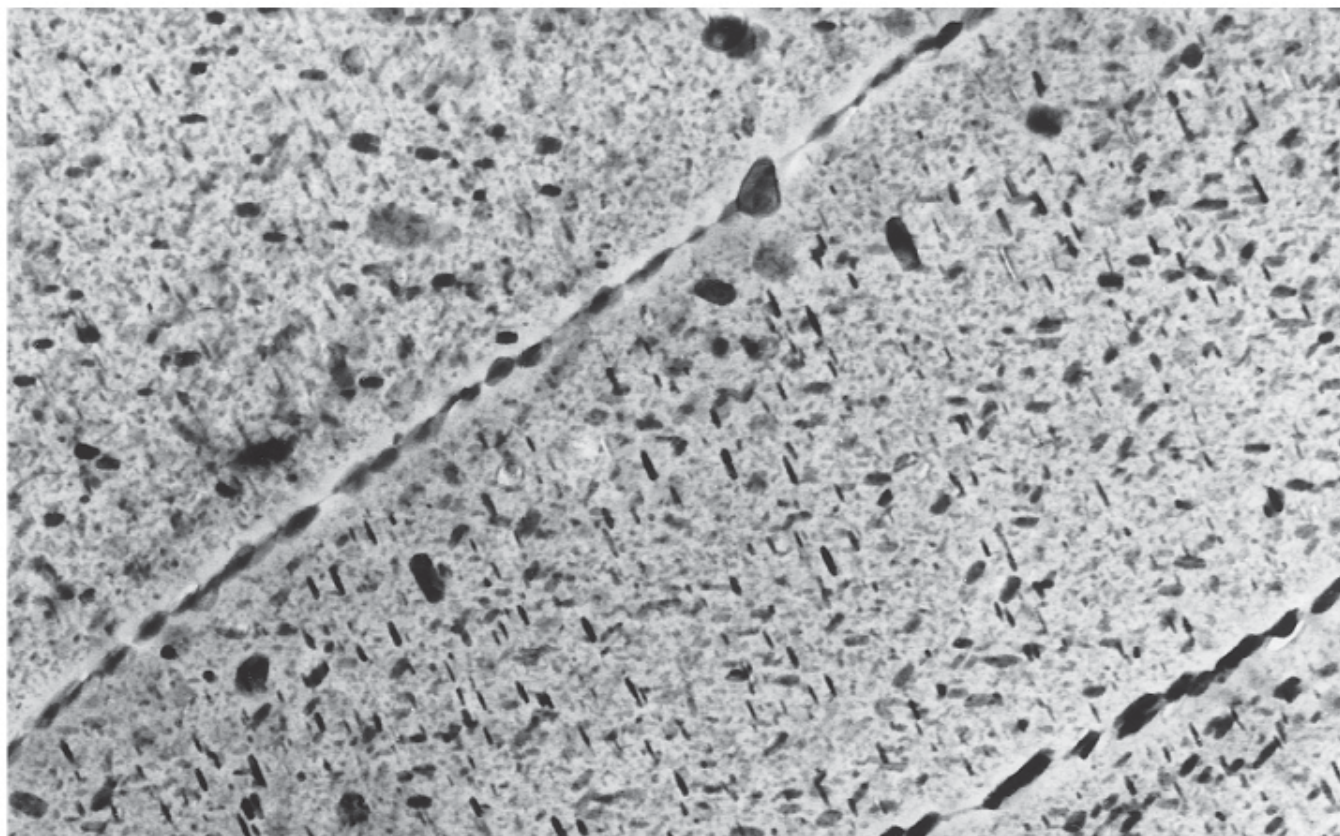


Figura 11.26 Micrografia elettronica a trasmissione che rappresenta la microstruttura della lega di alluminio 7150-T651 (6.2 Zn, 2.3 Cu, 2.3 Mg, 0.12 Zr, resto Al) che è stata indurita per precipitazione. La fase chiara nella micrografia è la matrice, una soluzione solida di alluminio. La maggior parte delle piccole particelle precipitate scure a forma di piccole piastrine sono una fase di transizione η' , il rimanente è la fase in equilibrio η (MgZn_2). Notare che i bordi dei grani sono “decorati” da alcune di queste particelle. 90000 $\times$. (Micrografia elettronica gentilmente concessa da G. H. Narayanan and A. G. Miller, Boeing Commercial Airplane Company).

Mechanical properties vs. microstructure of Al alloys and thermal treatments

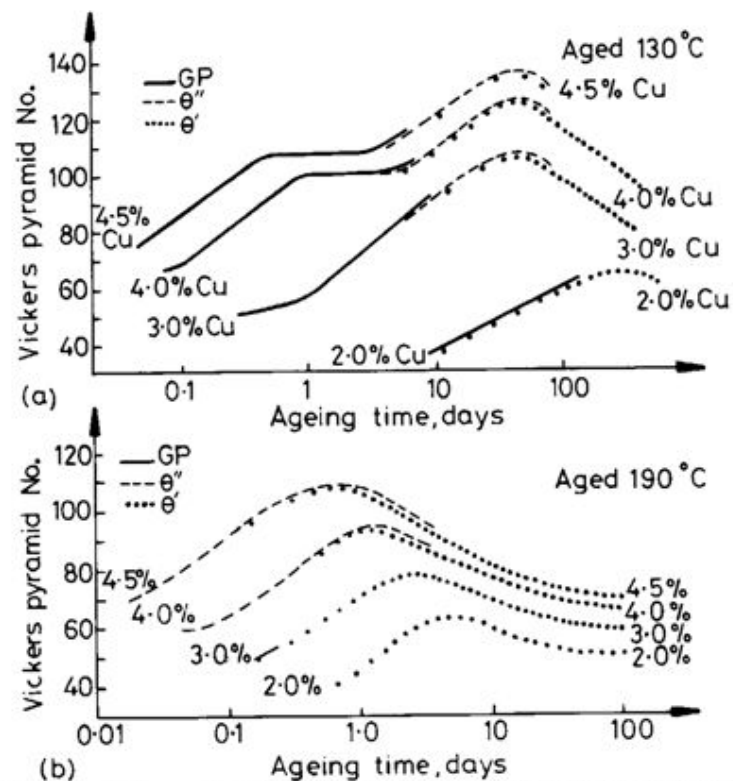
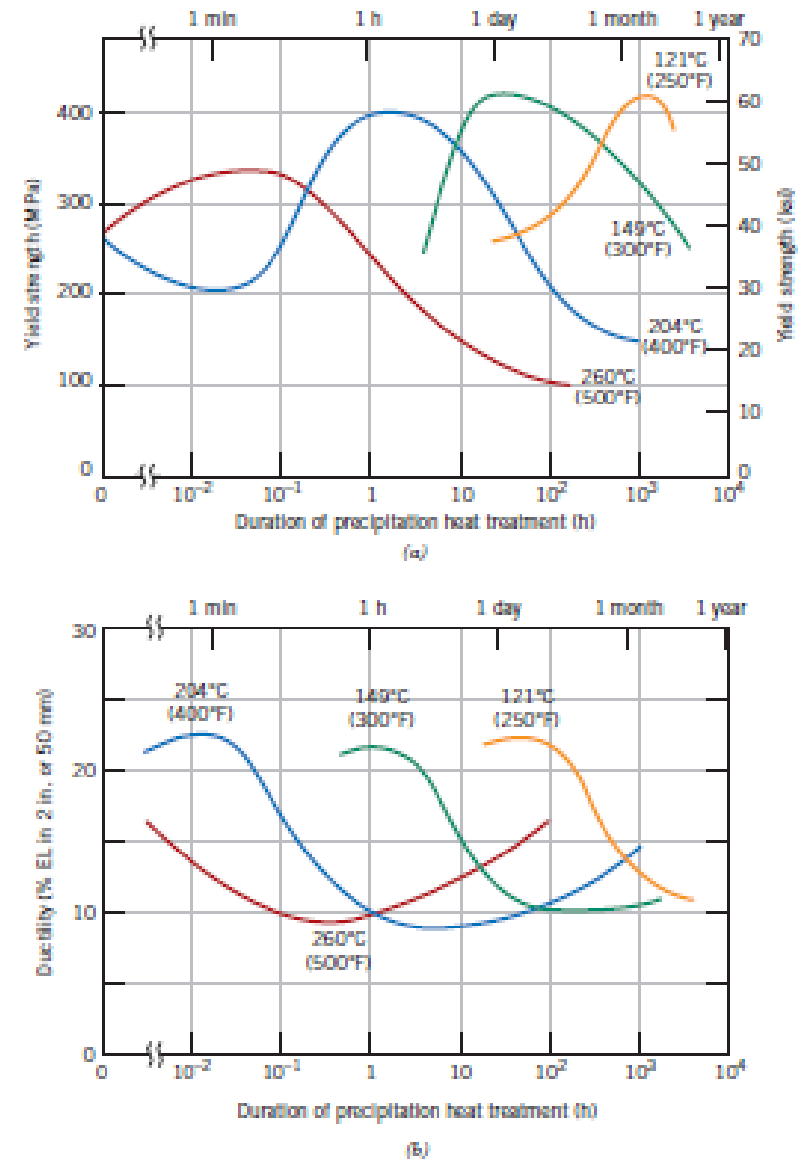


Fig. 5.37 Hardness v. time for various Al-Cu alloys at (a) 130 °C (b) 190 °C. (After J.M. Silcock, T.J. Heal and H.K. Hardy, *Journal of the Institute of Metals* 82 (1953-1954) 239.

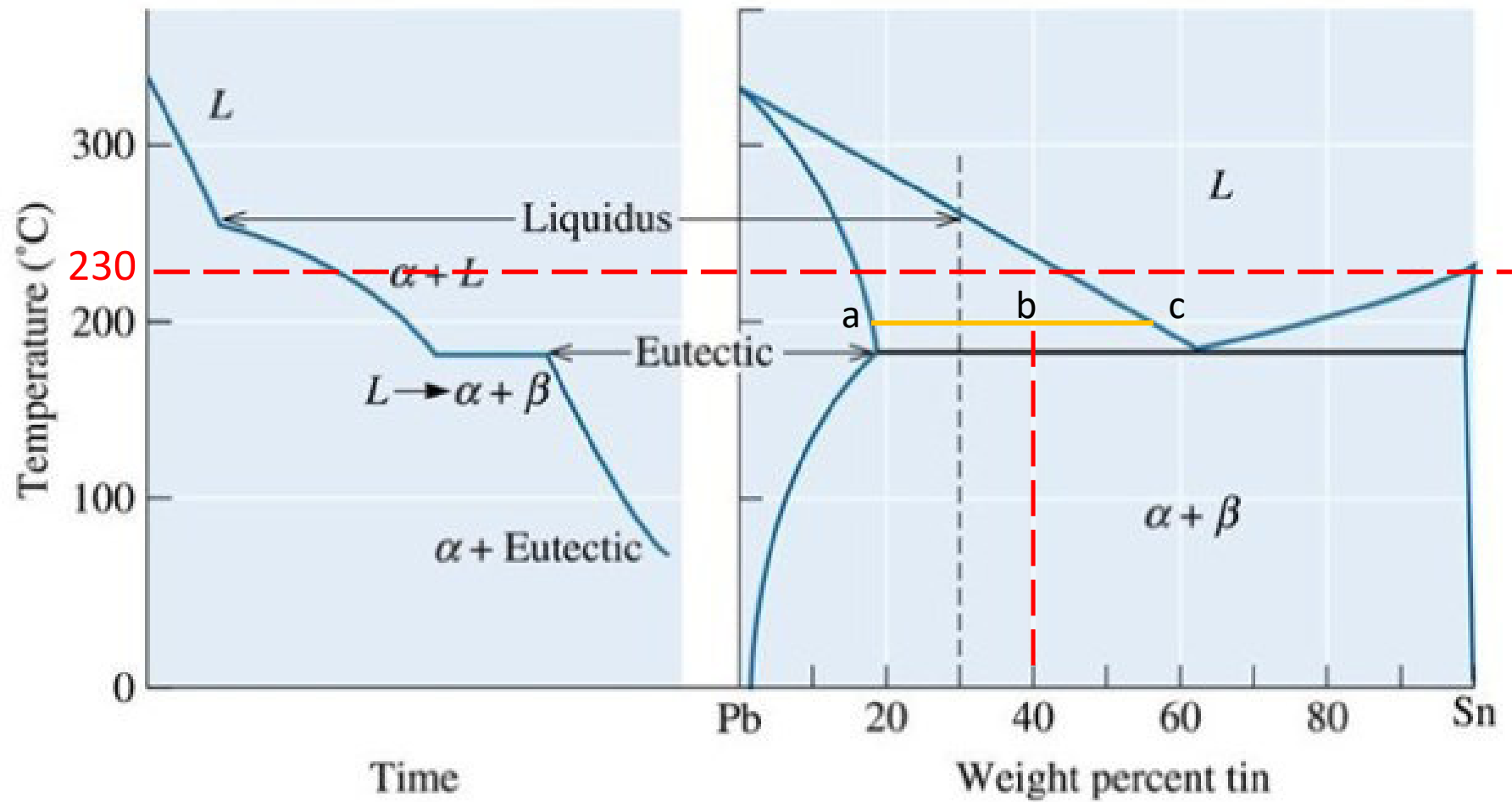
Figure 11.27 The precipitation hardening characteristics of a 2014 aluminum alloy (0.9 wt% Si, 4.4 wt% Cu, 0.8 wt% Mn, 0.5 wt% Mg) at four different aging temperatures: (a) yield strength, and (b) ductility (%EL). [Adapted from *Metals Handbook: Properties and Selection: Nonferrous Alloys and Pure Metals*, Vol. 2, 9th edition, H. Baker (Managing Editor), American Society for Metals, 1979, p. 41.]



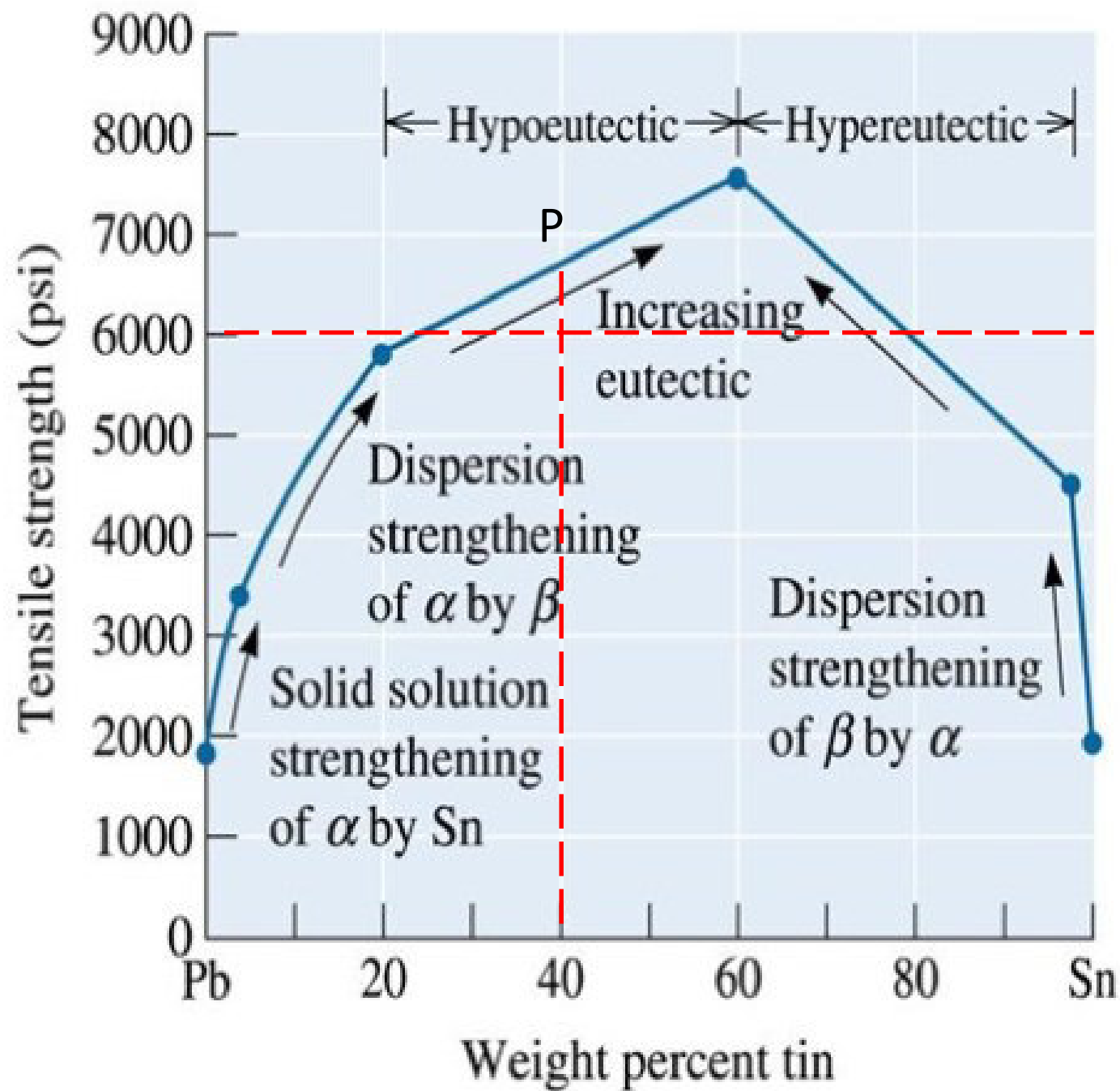
Higher amount of Cu → higher reinforcing effect → higher density of precipitates

Practical examples of the use of phase diagrams in materials design

One way to repair dents in a metal is to wipe a partly liquid-partly solid material into the dent, then allow this filler material to solidify. For our application, the wiping material should have the following specifications: (1) a melting temperature below 230°C , (2) a tensile strength in excess of 6000 psi, (3) be 60% to 70% liquid during application, and (4) the lowest possible cost. Design an alloy and repair procedure that will meet these specifications.



At 205 °C $\rightarrow (ab/ac) \times 100 = 65\%$



Our recommendation, therefore, is to use a Pb-40% Sn alloy applied at 205°C, a temperature at which there will be 65% liquid and 35% primary α . As mentioned before, we should also pay attention to the toxicity of lead and any legal liabilities the use of such materials may cause. A number of new lead free solders have been developed.

Historical example

In 1958, at the beginning of the US space project, an engineer proposed to use tungsten (W) with a purity of 99.9% as the material for the exhaust nozzles of the engines.

The choice was based on a few points:

- tungsten has a melting temperature of 3410 °C, about 400 °C above the temperature of the exhaust gases;
- tungsten is hard and can resist abrasion due to hot exhaust gases
- tungsten was not too expensive (\$ 25,000 per nozzle) as compared to advanced ceramics at that time

Three prototypes are produced and tests started.

The first nozzle only resisted a few seconds at about 3000 °C and then melted

The second and third nozzles melted even before reaching 3000 °C

The residues of the two destroyed nozzles were analyzed; the composition of the material was, in both cases, W = 98.5% and C = 1.5%

